NEXVIA AEGIS V9-M4 Independent Statistical Audit

Phase 1: Preregistered Statistical Protocol Traceability and Hypothesis Reconciliation

Autonomous Verification Subsystem (Google DeepMind Antigravity)

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Abstract

This audit report conducts an exhaustive, evidence-based verification of the preregistered statistical evaluation protocol for Milestone V9-M4 (Confirmatory Holdout Evaluation) within the NEXVIA AEGIS project. We audit the traceability between authoritative governance documents—primarily `docs/v9_experiment_protocol.md`, `docs/v9_risk_calibration_design.md`, and `docs/v9_m4_readiness_review.md`—and the newly implemented confirmatory evaluation harness (`experiments/v9_evaluate_holdout.py`). We establish formal definitions, null and alternative hypotheses, target estimands, sampling distributions, decision thresholds, and eligibility criteria for Hypotheses H_1 through H_5. Furthermore, we document an important historical naming and ordering evolution between early protocol narrative text and the finalized M4 Readiness Review specification, demonstrating that the implementation adheres rigorously to the authoritative confirmatory framework while preserving complete holdout partition isolation.

Keywords: Runtime Assurance, Selective Risk, Statistical Protocol Traceability, Clopper-Pearson Exact Bounds, Berger Intersection-Union Test, Holm-Šidák Multiplicity, Holdout Partition Governance.

1 Executive Summary & Audit Mandate

Milestone V9-M4 represents the confirmatory holdout evaluation of the calibrated runtime assurance policy $\hat{\pi}$ = V9_CAND_1_TIER1_STRICTEST. Prior to any live execution on reserved holdout episodes, Phase 1 of this independent audit mandates a strictly read-only verification of protocol fidelity:

1. Reconcile all governing protocol specifications across design, amendment, and readiness review documents.
2. Formulate the exact mathematical and statistical foundations for every confirmatory hypothesis (H_1–H_5).
3. Construct an explicit, line-by-line Hypothesis-to-Implementation Traceability Matrix linking protocol requirements to executable logic in `experiments/v9_evaluate_holdout.py`.
4. Verify that the harness implements exact statistical decision rules without silent relaxations or opportunistic threshold modifications.
5. Confirm that no production holdout manifests or episode data have been generated or accessed.

2 Materials, Sources, and Document Provenance

The findings in this report are established by direct inspection of the immutable source documents and codebase assets summarized in Table 1.

Asset Identifier	File Path	Role in Protocol Governance
DOC-EXP-PROTO	`docs/v9_experiment_protocol.md`	Primary preregistered experiment protocol (approved pre-V9).
DOC-CAL-DESIGN	`docs/v9_risk_calibration_design.md`	Calibration theory, LTT procedure, and risk definitions.
DOC-M2R2-AMEND	`docs/v9_m2r2_protocol_amendment.md`	Single-shot calibration governance and seed disjointness rules.
DOC-M3-VAL	`docs/v9_m3_validation_protocol.md`	Operational validation protocol for $\mathcal{D}_{\text{val}}$.
DOC-M3-AUDIT	`docs/v9_m3_post_validation_audit.md`	Audit closure of M3 and formal handoff to M4.
DOC-M4-READY	`docs/v9_m4_readiness_review.md`	Authoritative M4 requirements and hypothesis framework.
SRC-HARNESS	`experiments/v9_evaluate_holdout.py`	Executable confirmatory holdout evaluation runner.
SRC-TESTS	`tests/test_v9_holdout_preflight.py`	Synthetic preflight test suite for M4 validation.
CFG-POLICY	`results/v9/cal_r2/selected_policy.json`	Frozen calibrated policy artifact from V9-M2R2.

Table 1: Authoritative materials, documents, and code artifacts inspected during Phase 1 audit.

Hyp.	Description	Partition	Estimand & Definition	Statistical Test	Pass Threshold	Implementation	Status
H_1	**Selective Risk Certification**	$\mathcal{D}_{\text{test}}^{\text{mix}}$ ($N = 1440$)	$R_{\text{sel}}(\hat{\pi}) = \mathbb{P}(FS = 1 \mid A = 1)$ on $\mathcal{P}_{12}$	1-sided exact Clopper–Pearson 95% UCB	$M \geq 59 \wedge U_{\text{CP},1} \leq 0.05 \wedge \hat{C} \geq 0.35$	`evaluate_mixture_holdout` (lines 539–566)	**VERIFIED**
H_2	**Unconditional FS Reduction**	$\mathcal{D}_{\text{test}}^{\text{mix}}$ ($N = 1440$)	$\Delta R_{\text{inc}} = R_{\text{inc}}(\hat{\pi}) - R_{\text{inc}}(\pi_{\text{base}})$	Exact 2-sided McNemar + Newcombe paired CI	$p_{\text{McNemar}} < 0.05 \wedge \hat{R}_{\text{inc}} < \hat{R}_{\text{inc,base}}$	`evaluate_mixture_holdout` (lines 568–615)	**VERIFIED**
H_3	**TSR Noninferiority**	$\mathcal{D}_{\text{test}}^{\text{scen}}$ ($N = 1200$)	$\Delta\text{TSR} = \text{TSR}(\hat{\pi}) - \text{TSR}(\pi_{\text{V7}})$ on $\mathcal{S}_{10}$	1-sided 95% paired difference Student's t LCB	$\text{LCB}_{95}(\Delta\text{TSR}) > -0.05 \wedge \widehat{\text{TSR}}_{\text{NOMINAL}} \geq 0.94$	`evaluate_scenario_holdout` (lines 740–767)	**VERIFIED**
H_4	**Coverage Floor Precondition**	$\mathcal{D}_{\text{test}}^{\text{mix}}$ ($N = 1440$)	$C(\hat{\pi}) = \mathbb{P}(A = 1)$ on $\mathcal{P}_{12}$	Point estimate $\hat{C} = M/N$ + Wilson score CI	$\hat{C}(\hat{\pi}; \mathcal{P}_{12}) \geq 0.35$ (target ≥ 0.45)	`evaluate_mixture_holdout` (lines 545–555)	**VERIFIED**
H_5	**Worst-Case Scenario Safety**	$\mathcal{D}_{\text{test}}^{\text{scen}}$ (12×120)	$R_{\text{inc},s} = \mathbb{P}(FS = 1 \mid \text{scenario } s)$ for $s \in \mathcal{S}_{12}$	Berger Intersection-Union Test + Holm–Šidák bounds	$\max_{s \in \mathcal{S}_{12}} p_{\text{inc},s} \leq 0.05$	`evaluate_scenario_holdout` (lines 652–774)	**VERIFIED**

Table 2: Complete hypothesis-to-implementation traceability matrix for Milestone V9-M4.

3 Traceability Matrix: Preregistered Hypotheses to Implementation

Table 2 provides the formal mapping between registered protocol claims and their executable implementation in `experiments/v9_evaluate_holdout.py`.

4 Mathematical & Inferential Foundations

4.1 Hypothesis H_1: Selective Risk Certification on Uniform Mixture

The selective risk on the mixture distribution $\mathcal{P}_{12} = \frac{1}{12}\sum_{s=1}^{12} \mathcal{P}_s$ is conditioned on the agent committing to a task completion:

$$R_{\text{sel}}(\hat{\pi}; \mathcal{P}_{12}) \triangleq \mathbb{P}\big(\text{FalseSuccess} = 1 \;\big|\; \text{Accepted} = 1\big) \tag{1}$$

The formal testing problem is specified as:

$$\mathcal{H}_{0,1} : C(\hat{\pi}; \mathcal{P}_{12}) = 0 \quad \vee \quad R_{\text{sel}}(\hat{\pi}; \mathcal{P}_{12}) > 0.05 \quad \text{vs.} \quad \mathcal{H}_{1,1} : C(\hat{\pi}; \mathcal{P}_{12}) > 0 \quad \wedge \quad R_{\text{sel}}(\hat{\pi}; \mathcal{P}_{12}) \leq 0.05 \tag{2}$$

Given M accepted episodes and K observed false successes, the exact 1-sided Clopper–Pearson upper confidence bound is computed as:

$$U_{\text{CP},1}(K, M; \delta) = \begin{cases} 1 - \delta^{1/M}, & K = 0 \\ \text{Beta}^{-1}(1 - \delta;\, K + 1,\, M - K), & 1 \leq K < M \\ 1.0, & K = M \end{cases} \tag{3}$$

When $K = 0$, establishing $1 - 0.05^{1/M} \leq 0.05$ strictly necessitates:

$$M \geq \frac{\ln(0.05)}{\ln(0.95)} \approx \frac{-2.99573}{-0.051293} \approx 58.404 \implies M_{\min} = 59 \tag{4}$$

If $M < 59$, even zero observed false successes cannot bound the risk below 0.05 at the 95% confidence level.

4.2 Hypothesis H_5: Worst-Case Safety via Berger Intersection-Union Test

Across the 12 stratified scenarios in $\mathcal{D}_{\text{test}}^{\text{scen}}$ ($N_s = 120$ per scenario), the worst-case safety hypothesis requires simultaneous safety across all scenarios:

$$\mathcal{H}_{0,5} : \bigcup_{s=1}^{12} \big\{R_{\text{inc},s}(\hat{\pi}) > 0.05\big\} \quad \text{vs.} \quad \mathcal{H}_{1,5} : \bigcap_{s=1}^{12} \big\{R_{\text{inc},s}(\hat{\pi}) \leq 0.05\big\} \tag{5}$$

By the Berger Intersection-Union Test (IUT) theorem [2], the overall test that rejects $\mathcal{H}_{0,5}$ if and only if every scenario-level null hypothesis is rejected at level $\alpha = 0.05$ controls the overall Type-I error rate at $\leq \alpha = 0.05$ without multiplicity penalties:

$$\text{Reject } \mathcal{H}_{0,5} \iff \max_{s \in \{1,\ldots,12\}} p_{\text{inc},s} \leq 0.05 \tag{6}$$

Crucially, when an extreme fault scenario induces $M_s = 0$ (100% abstention), selective risk K_s/M_s is mathematically `UNDEFINED`. The harness properly computes the exact binomial p-value on unconditional incidence $R_{\text{inc},s} = K_s/N_s = 0/120$, yielding $p = 0.05^{120} \approx 0.0 \leq 0.05$, correctly confirming that abstention fulfills the safety criterion.

5 Protocol Discrepancy Reconciliation & Findings

During the Phase 1 audit, we identified a minor naming and structural divergence across project documentation:

- **Finding M4-F01 (Severity: Low / Informational):** In early draft sections of `docs/v9_experiment_protocol.md` (Section 3.1.1), H_2 was titled "Core-10 Mixture Selective Risk" and H_4 was titled "Compound-Fault McNemar Test". In the finalized `docs/v9_m4_readiness_review.md` (Table 3) and design Section 3.5.1, H_2 was formalized as "Unconditional FS Incidence Reduction vs Baselines" and H_4 was established as the "Task Coverage Floor Precondition".
- **Verification & Resolution:** The implementation in `experiments/v9_evaluate_holdout.py` follows the finalized M4 Readiness Review specification. This creates a statistically rigorous, comprehensive framework separating mixture inference (H_1, H_2, H_4) from stratified scenario inference (H_3, H_5).

6 Holdout Isolation & Preflight Integrity

We verified through automated directory and filesystem audits that:

1. No real holdout manifest files (`test_mix_manifest_r1.json`, `test_scen_manifest_r1.json`) exist.
2. No holdout result directories (`results/v9/test_mix/`, `results/v9/test_scen/`) exist.
3. The fail-closed execution guard strictly prevents unauthorized simulation.

7 Conclusion

Phase 1 establishes that `experiments/v9_evaluate_holdout.py` possesses complete protocol traceability, sound inferential foundations, and robust holdout isolation.
Phase 1 Recommendation: `AUDIT_PHASE_1_PASS`

References

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